Fiscal Federalism and Monetary Unions

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The Question of Delegation

- How should policy choices be delegated between central and local fiscal authorities?

- Answer important for many countries, for instance
  - ongoing debate in the European Union about fiscal integration
    - most fiscal decisions are determined at the country-level although subject to some EU-level rules
    - recent agreement towards a more gradual and tailored fiscal adjustment
    - and new policies hinting towards more fiscal integration: NextGenEU, SURE
  - also relevant for fiscal delegation at the country-level
    - complex rules of fiscal federalism in Argentina responsible affect its economic and policy performance (Saiegh and Tommasi 1999, Nicolini et al. 2002, Cooper and Kempf 2004)

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- Next: what are the answers in the literature?

Answers From Literature

- Macro lit. on monetary unions: centralized fiscal authority is *always weakly better* (Chari and Kehoe, 2008, Aguiar et al, 2015)
  - presumes that absent externalities local and central are equally good
  - so even *tiny* externalities make centralized better because it internalizes them
  - idea: if a country increases nominal debt it induces mon. authority to increase inflation
  - high inflation is costly for all countries: central authority internalizes these costs

→ main takeaway from most of this literature: *no benefit* to decentralized authority

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- Micro lit. on fiscal federalism: local authority is better unless fiscal externalities are fairly high
  (Oates 1972)
    - idea: local authorities are better at tailoring policies to the tastes of local citizens

→ main takeaway from this literature: in general *large benefits* to decentralized authority

Our Approach to the Benefits of Centralization vs. Decentralization

- This paper: incorporate two key forces
  - Benefit of decentralization in the spirit of fiscal federalism literature
    - central authority observes only noisy signal of local preferences
    - information problem prevents central authority to elicit them
  - Benefit of centralization in the spirit of the macro literature
- Dynamic model: captures how debt *dynamics* in union vary depending on fiscal regime
  - central fiscal authority internalizes the inflationary cost of debt

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  - Benefit of centralization in the spirit of the macro literature
- Dynamic model: captures how debt *dynamics* in union vary depending on fiscal regime
  - central fiscal authority internalizes the inflationary cost of debt
- Main goal: characterize when is it optimal to centralize fiscal authority
- Main result:
  - as the number of countries in the union expands, centralization becomes more desirable

Model Set-Up

- Incorporate strategic interactions (finite countries, I) and information friction to Aguiar et al (2015)
- Each region/country in the monetary union has a representative agent
- All countries are identical except for their preferences between public and private goods
  - we abstract from transfers across countries or any redistribution mechanism
- Compare two regimes: local vs. central fiscal authority (decentralized vs. centralized)
- Either local or central authority chooses nominal debt issued to foreign lenders
- Linear production function using labor: $y_{it} = \ell_{it}$ with $\ell_{it} \in [0, \overline{\ell}]$

Preferences and Information Structure

- The representative agent in each country
  - gets utility from private consumption, c, and public goods, g
  - linear disutility from working, and direct disutility from inflation, $\psi\pi$
- So, preferences in country i are given by

$$\mathbb{E}\int_0^\infty e^{-\rho t}\left[(1-\theta_{it})u\left(c_{it}\right)+\theta_{it}h(g_{it})-\ell_{it}-\psi\pi_t\right]dt$$

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- Suppose θ are iid shocks across countries
- Local fiscal authority: perfectly observes θ_i
- Central fiscal authority: observes noisy signal s_i about it
- Idea: local authority tries to communicate θ_i but this type of communication difficult
  - e.g. preferences over complex policies are nearly impossible to fully specify

Foreign Lenders and Debt Dynamics

- Risk-neutral foreign lenders buy non-defaultable government bonds, b_{it} (in real units)
- Their real opportunity cost is ρ which equals the discount rate of consumers
- The law of motion of debt in country i is

$$\dot{b}_{it} = c_{it} + g_{it} + (i_t - \pi_t)\, b_{it} - \ell_{it}$$

were i_t is the nominal interest rate

Monetary Authority

- The union-wide monetary authority maximizes utility of all countries in union by choosing inflation
- If the monetary authority has commitment, then there is no benefit to inflation as real interest rate is

$$i_t - \pi_t = \rho$$

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  - in this case, monetary authority chooses $\pi_t = 0$ for all t
- We study the interesting case: no commitment
  - temptation to inflate: decreases the real value of debt to be repaid
  - cost of inflation: direct linear utility cost: $\psi \pi_t$
  - in equilibrium, of course, the real interest rate is ρ but π is positive if debt is high enough

Monetary Authority: How Does It Choose Inflation?

- The union-wide monetary authority maximizes utility of all countries in the union
- Given a vector of current debt in each country $\boldsymbol{b} = (b_1, \ldots, b_I)$ and preferences $\boldsymbol{\theta}$, chooses inflation
- Assume that $\pi_t \in [0, \overline{\pi}]$. So, it solves

$$J(\boldsymbol{b}_0, \boldsymbol{\theta}_0) = \max_{\{\pi_t\}} \frac{1}{I} \sum_i \mathbb{E}_0 \int_0^\infty e^{-\rho t} [(1-\theta_{it}) u(c_{it}) + \theta_{it} h(g_{it}) - \ell_{it} - \psi \pi_t] dt$$

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- Formulate as an HJB equation and show that optimal inflation is given by

$$\pi(\boldsymbol{b}, \boldsymbol{\theta}) \begin{cases} = 0 & \text{if} \quad \psi > -\sum_i \frac{\partial J(\boldsymbol{b},\boldsymbol{\theta})}{\partial b_i} b_i \\ \in [0, \bar{\pi}] & \text{if} \quad \psi = -\sum_i \frac{\partial J(\boldsymbol{b},\boldsymbol{\theta})}{\partial b_i} b_i \\ = \bar{\pi} & \text{if} \quad \psi < -\sum_i \frac{\partial J(\boldsymbol{b},\boldsymbol{\theta})}{\partial b_i} b_i \end{cases}$$

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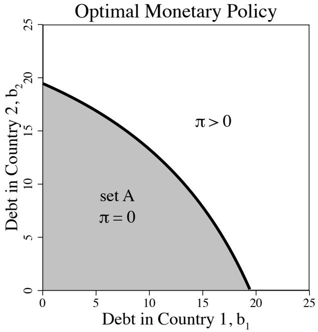
$\rightarrow$ Next, fiscal authorities problem: proceed by steps

Intuition for $I = 2$ With No Information Problem: $\theta = 0$ and $g = 0$

- The optimal inflation rule is of the following form

$$\pi(b_1, b_2) = \begin{cases} 0 & \text{if} \quad (b_1, b_2) \in A \\ \bar{\pi} & \text{if} \quad (b_1, b_2) \in A^C \end{cases}$$



Intuition for $I = 2$: Three Regions Depending on Debt

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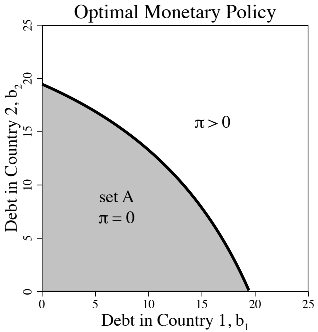

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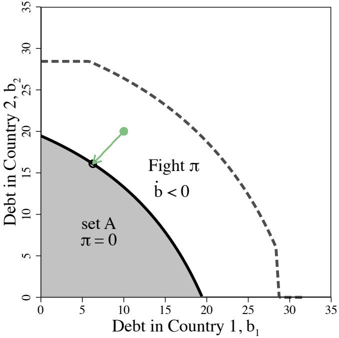

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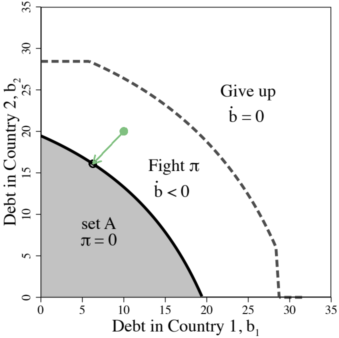

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- **Key**: in a centralized regime *fight inflation* at higher debt levels than decentralized
  - and decrease debt faster

$\rightarrow$ Next: formally show these results, starting with centralized fiscal authority problem

The Problem of the Centralized Fiscal Authority

- Focus on the symmetric case for now: same initial debt in each country, $b_1 = \cdots = b_I = b$
- Central fiscal authority chooses cons. and labor in each country (Ramsey problem within country)
- Maximizes the equally weighted sum of utilities in the union
- Easy to show that it chooses same allocation in each country (drop index i)
- Taking as given $i(b)$ and $\pi(b)$, the value of the centralized fiscal authority is then

$$V^C(b) = \max_{c_t, \ell_t \in [0, \bar{\ell}]} \int_0^\infty e^{-\rho t} \left[u(c_t) - \ell_t - \psi \pi(b_t) \right] dt$$

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- So, the HJB is given by

$$\rho V^C(b) = \max_{c, \ell \in [0, \overline{\ell}]} \quad u(c) - \ell - \psi \pi(b) + \frac{\partial V^C(b)}{\partial b} (c - \ell + \rho b)$$

Centralized Fiscal Authority: Markov Perfect Equilibrium

- Focus on best monotone Markov Perfect Equilibrium and symmetric case: $b_1 = \cdots = b_I = b$

- An equilibrium is
  - interest rate schedule, $i(b) = \rho + \pi(b)$,
  - fiscal policy functions, $c(b)$ and $\ell(b)$,
  - inflation rule, $\pi(b)$,
  - and, value functions for the monetary authority, $J(b)$, and fiscal authority $V^C(b)$

  such that,

  i) given interest rate schedule and inflation rule, the fiscal authority maximizes utility

  ii) given interest rate schedule and fiscal rules, the monetary authority maximizes utility

- Note that from the definition, $J(b) = V^C(b)$

Centralized Fiscal Authority: Characterization of Equilibrium

- Inflation rule from monetary authority problem is (as in motivating figure)

$$\pi(b) = \begin{cases} 0 & \text{if} \quad \psi \geq -J'(b)b \\ \bar{\pi} & \text{if} \quad \psi < -J'(b)b \end{cases}$$

  where $J(b)$ is the value of function of the monetary authority

  - compare marginal cost of inflation ψ with its marginal benefit

- Note that with no inflation, $\pi = 0$, if solution is interior, fiscal authority chooses consumption s.t.

  - marginal utility of c equals marginal disutility of working (linear): $u'(c^*) = 1$

- Assume that the upper bound of labor is such that $\bar{\ell} > c^* + \rho b$

  - ensures that if inflation is equal to zero, then the optimal allocation of labor is interior

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$$\pi(b) = \begin{cases} 0 & \text{if} \quad \psi \geq b \\ \bar{\pi} & \text{if} \quad \psi < b \end{cases}$$

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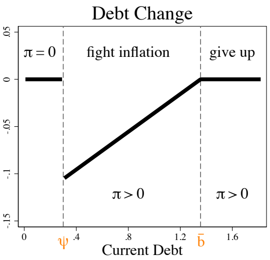

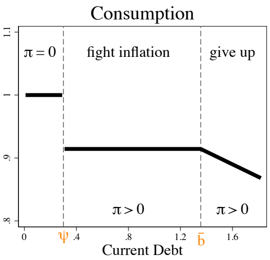

1. *Zero inflation*: equilibrium solution is $u'(c^*) = 1$ and $\ell = c^* + \rho b < \overline{\ell}$

2. *Fight inflation*: countries decrease their debt level to fight inflation
   - and choose a constant level of consumption along the path of debt reduction, $\underline{c}$

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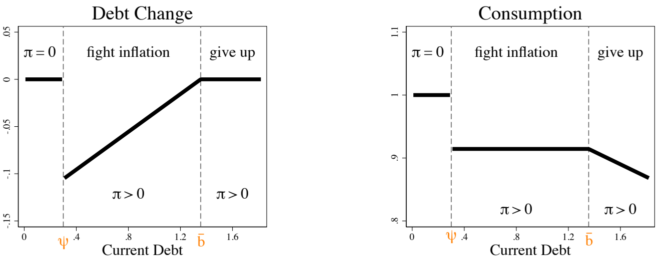

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$\rightarrow$ Note: all the results in the centralized case are independent of I; next, decentralized fiscal authority

The Problem of the Local Fiscal Authority

- Each country's local fiscal authority chooses (c_i, ℓ_i) to maximize utility of their country
- Strategic interaction: inflation depends on what all other countries are doing
  - need to take other countries' debt evolution into account
- Given $\pi(\boldsymbol{b}), i(\boldsymbol{b})$, and $\{c_j(\boldsymbol{b}), \ell(\boldsymbol{b})\}_{j \neq i}$, the local fiscal authority value is given by

$$V_i^D(\boldsymbol{b}_0) = \max_{c_{it}, l_{it} \in [0, \bar{\ell}]} \int_0^\infty e^{-\rho t} \left[u(c_{it}) - \ell_{it} - \psi \pi(\boldsymbol{b}_t) \right] dt$$

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Local Fiscal Authority: Characterization of Equilibrium

- The optimal inflation rule is

$$\pi(\boldsymbol{b}) = \begin{cases} 0 & \text{if} \quad \psi \geq -\sum_i \frac{\partial J(\boldsymbol{b})}{\partial b_i} b_i \\ \bar{\pi} & \text{if} \quad \psi < -\sum_i \frac{\partial J(\boldsymbol{b})}{\partial b_i} b_i \end{cases}$$

  where $J(\boldsymbol{b})$ is the value of the monetary authority

- Since $J(\boldsymbol{b}) = \frac{1}{I} \sum_i V_i^D(\boldsymbol{b})$, we can show that

$$-\sum_i \frac{\partial J(\boldsymbol{b})}{\partial b_i} = -\frac{1}{I} \sum_i \frac{\partial V_i^D(\boldsymbol{b})}{\partial b_i} = \frac{1}{I} \sum_i u'(\tilde{c}_i^*(\boldsymbol{b}))$$

  where $\tilde{c}_i^*(\boldsymbol{b}) = \min\{c^*, \bar{\ell} - \rho b\}$ is optimal consumption with c^* s.t. $u'(c^*) = 1$ is interior solution

Local Fiscal Authority: Characterization of Equilibrium

- **Proposition:** The optimal inflation rule is

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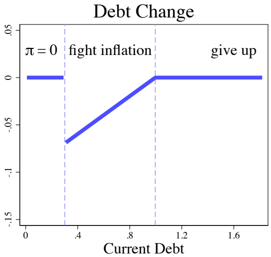
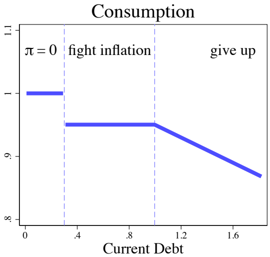
- Similar to the case with a centralized fiscal authority
  - but now, need to contemplate the possibility of deviating from symmetric equilibrium
  - take into account the effect of changing b_i in each country

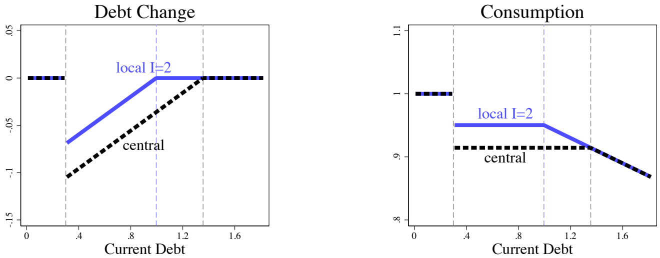
Local Fiscal Authority: Characterization of Equilibrium with $I = 2$





- Same qualitative form as in the centralized regime
- But both consumption level and area where countries fight inflation depend on no. of countries I
- Next, compare with the centralized regime

Compare Local and Central Fiscal Authority Equilibria with $I = 2$



1. *Zero inflation:* equilibrium same in both regimes

2. *Fight inflation*:
   - ∘ in both regimes, consumption is constant along the debt reduction path, but $\underline{c}^D(I) > \underline{c}^C$
   - ∘ debt decreases slower in the decentralized so takes longer to get zero inflation

   → key fiscal externality: locals don't internalize the union-wide benefits of decreasing debt fast

3. *Give up fighting inflation*: for lower levels of debt under decentralized

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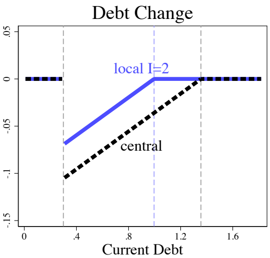
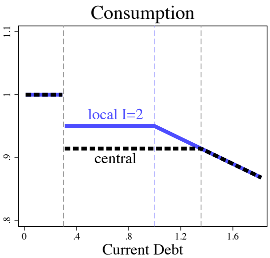
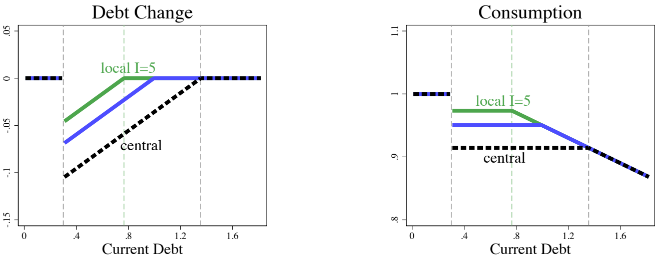




What about welfare?

- When inflation is zero or both give up inflation: same allocations and welfare in two regimes
- In the area where centralized regime fights inflation:
    - flow utility is higher in decentralized because consumption is higher
    - but, overall welfare higher under centralized because it gets to $\pi = 0$ and faster

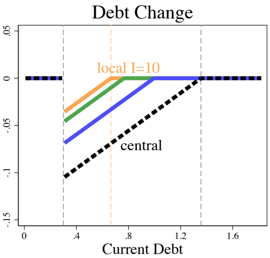
Compare Local and Central Fiscal Authority Equilibria with $I = 5$

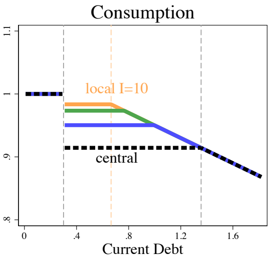


- As the number of countries in the union I increase
  - don't *fight inflation* as hard: $\underline{c}^D(I)$ increases with I
  - so the rate at which debt decreases is slower: takes longer to reach the zero inflation area
  - *give up* fighting inflation for lower levels of debt

$\rightarrow$ fiscal externality becomes worse

Compare Local and Central Fiscal Authority Equilibria with $I = 10$





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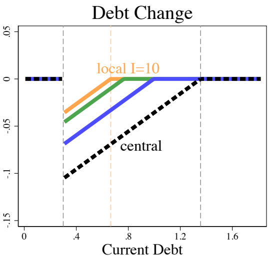
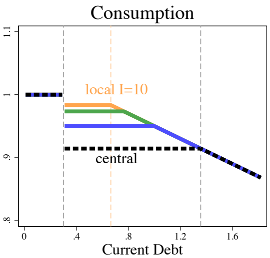




Proposition: in the symmetric case $b_1 = \cdots = b_I$

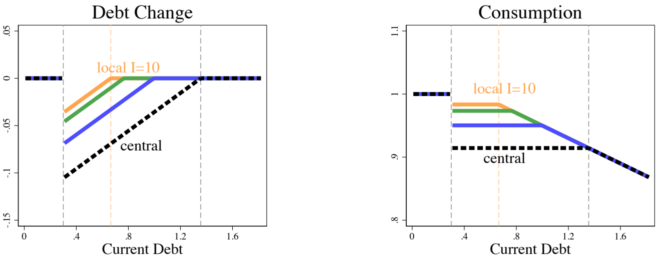
i) if $b \leq \psi$ (*no inflation*) or $b \geq \overline{b}(I = 1)$ (*giving up* under centralized)

- a decentralized regime is as good as a centralized one

ii) if $b \in (\psi, \overline{b})$ (when *fighting inflation*), then a centralized regime is preferred

- and the value of a decentralized regime decreases with I for $b \in (\psi, \overline{b}(I))$

Compare Local and Central Fiscal Authority Equilibria



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Let $J^C(\boldsymbol{b})$ and $J^D(\boldsymbol{b}, I)$ denote the ex-ante welfare under central and local regimes in this problem

Next: add an information disadvantage to the centralized regime

Add an Information Disadvantage to Central Fiscal Authority

- Go back to the general problem in which countries have heterogeneous preferences about g

- Preferences in each country i are given by

$$\mathbb{E}\int_0^\infty e^{-\rho t}\left[(1-\theta_{it})u\left(c_{it}\right)+\theta_{it}h(g_{it})-\ell_{it}-\psi\pi_t\right]dt$$

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- Information structure formally
  - let $\boldsymbol{\theta_t} \equiv (\theta_{1t},\ldots,\theta_{It})$ be a random variable in probability space $(\Omega,\mathcal{F},\mathcal{P})$ and iid across i
  - local fiscal authority observes θ_{it} and its information structure is the filtration $\mathcal{F}_t^i=\sigma(\theta_{i\tau},\tau\leq t)$
  - central authority only observes signals $\boldsymbol{s_t}$ and info structure is filtration $\mathcal{F}_t^C=\sigma(\boldsymbol{s_\tau},\tau\leq t)$

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- An example: let $\theta_{it}\in\{\theta_L,\theta_H\}$ with $0<\theta_L<\theta_H<1$
    - at a given Poisson rate λ, preference θ_{it} switches from θ_L to θ_H and vice versa
    - central fiscal authority learns value of current θ_{it} with Poisson rate ϕ

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A Separation Result With Log Utility

Two parts to this separation result

- Debt dynamics identical to the economy with only fiscal externalities
  - total consumption, $c + g$, does not vary with θ, only its composition

- Welfare is sum of welfare with only externality and a term that captures benefits of info structure

A Separation Result With Log Utility

Proposition: The ex-ante welfare in an economy with heterogeneous preferences for g given by $\boldsymbol{\theta_t}$ is

$$\tilde{J}^C(\boldsymbol{b},\boldsymbol{\theta}) = J^C(\boldsymbol{b}) + f(\boldsymbol{\theta}|\mathcal{F}^C)$$
$$\tilde{J}^D(\boldsymbol{b},\boldsymbol{\theta},I) = J^D(\boldsymbol{b},I) + f(\boldsymbol{\theta}|\cap_i \mathcal{F}^i),$$

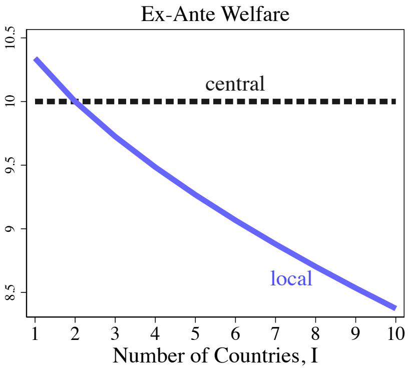
with $\hat{\theta}_{i,t} \equiv \mathbb{E}\left[\theta_{i,t}|\mathcal{F}_t\right]$, $\mathcal{F} = (\mathcal{F}_t)$, and

$$f(\boldsymbol{\theta}|\mathcal{F}) \equiv \frac{1}{I}\sum_i \mathbb{E}_{\boldsymbol{\theta}} \int_0^\infty e^{-\rho t}\left[\hat{\theta}_{i,t}\log\hat{\theta}_{i,t} + (1-\hat{\theta}_{i,t})\log(1-\hat{\theta}_{i,t})\right] dt,$$

where $J^C(\boldsymbol{b})$ and $J^D(\boldsymbol{b},I)$ are the value functions from the economy with only externalities

A Cutoff Rule Result

- There exists a cutoff in the number of countries $I(b; \mathcal{F}^C)$ s.t.
  - if I is small decentralization is preferred because of the info advantage
  - if I is large centralization is preferred because the externality becomes worse



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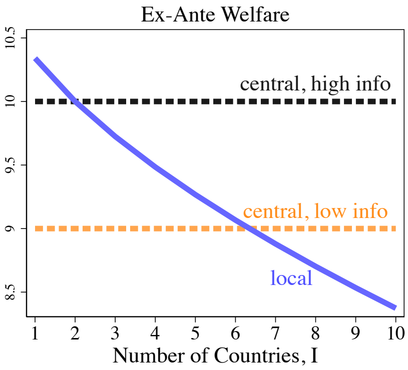
  - as information becomes worse, centralized welfare decreases, so cutoff increases



A Cutoff Rule Result

Proposition: Suppose that $(b_{i0}, \theta_{i0}) = (b, \theta)$ for all i and $\cap_i \mathcal{F}^i$ is strictly more informative than $\mathcal{F}^C$.

i) if $b \leq \psi$ (*no inflation*) or $b \geq \overline{b}(I = 1)$ (*give up* under centralized)

- then a decentralized regime is always preferred

ii) if $b \in (\psi, \overline{b})$, then a centralized regime is preferred if and only if $I > I(b; \mathcal{F}^C)$. Furthermore,

- $J^D(b, \theta, I + 1) < J^D(b, \theta, I)$ for $b \in (\psi, \overline{b}(I))$ (when *fighting inflation*)
- $J^D(b, \theta, I + 1) = J^D(b, \theta, I)$ for $b \in (\overline{b}(I), \overline{b})$ (when *give up*)

iii) the cutoff $I(\boldsymbol{b}; \mathcal{F}^C)$ decreases in the informativeness of $\mathcal{F}^C$: if $\mathcal{F}^C \subset \tilde{\mathcal{F}}^C$, then $I(\boldsymbol{b}; \mathcal{F}^C) \leq I(\boldsymbol{b}; \tilde{\mathcal{F}}^C)$

Conclusion

- Show how insights from fiscal federalism change principles of delegation from existing macro lit.
  - optimal delegation does not just depend on whether externalities exist or not
  - instead it depends on the trade-off between externalities and natural advantage of local authorities
- Implications for design of monetary union
  - more sophisticated trade-offs than in current macro literature
  - the dynamics of debt and transition paths depend on the fiscal regime in the union
  - key new idea: *centralization optimal only if monetary union sufficiently large*
  - analysis has implications for fiscal adjustment and enlargement policies